

Notes: Caliendo, Dvorkin, and Parro (Econometrica, 2019)

Trade and Labor Market Dynamics: General Equilibrium Analysis of the China Trade Shock

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1 Big picture

- **Question.** The paper asks how the rise of China in world trade affected U.S. labor markets once one accounts for general equilibrium, input-output linkages, internal geography, trade costs, and slow worker mobility across sectors and states.
- **Setting.** The empirical application is the China trade shock from 2000 to 2007. The authors study how increased Chinese competition changed U.S. employment, real wages, welfare, and adjustment costs across more than 1,000 state-sector labor markets.
- **Core modeling idea.** The paper combines an Eaton-Kortum style multisector trade model with a dynamic discrete-choice labor-mobility model in the spirit of Artuc, Chaudhuri, and McLaren. Workers are forward-looking and move gradually because switching sectors and states is costly.
- **Spatial and sectoral structure.** The quantitative model has 38 countries, 50 U.S. states, and 22 sectors. A U.S. labor market is a state-sector pair, plus non-employment. This creates 1,150 U.S. labor markets.
- **Methodological contribution.** The paper develops *dynamic hat algebra*: a way to solve the dynamic model in changes, relative to observed allocations, without estimating the level of productivities, trade costs, migration costs, local factor endowments, or home production values.
- **China shock measurement.** The China shock is measured as the sectoral Chinese productivity changes that make the model reproduce the Autor-Dorn-Hanson-style predicted increase in U.S. imports from China, using import growth from China in other high-income countries as the source of exogenous variation.
- **Main employment result.** The China shock reduces the U.S. manufacturing employment share by about 0.36 percentage points in the long run. This corresponds to about 0.55 million manufacturing jobs, around 16% of the observed 2000-2007 manufacturing employment decline.
- **Reallocation result.** Manufacturing jobs fall, but some workers move into services, wholesale and retail, and construction. The model estimates roughly 50,000 additional construction jobs due to cheaper Chinese intermediate inputs and reallocation.
- **Welfare result.** Aggregate U.S. welfare rises by about 0.2%, but the effects are highly heterogeneous across labor markets. Manufacturing labor markets are more exposed to direct competition, while many non-manufacturing markets benefit from cheaper inputs.
- **Dynamic result.** Long-run gains do not arrive immediately. Mobility frictions imply short-run real-wage losses in manufacturing regions and nontrivial transition costs. Accounting for adjustment dynamics is essential for welfare analysis.

2 Incremental contribution of the paper

- **From reduced-form exposure effects to general equilibrium levels.** Autor-Dorn-Hanson-type regressions identify relative employment losses across more- versus less-exposed commuting zones. This paper computes aggregate employment changes, sectoral spillovers, welfare effects, and policy counterfactuals.

- **Dynamic rather than static trade adjustment.** Static trade models can compare equilibria before and after shocks, but they miss the transition path. This paper explicitly models the gradual movement of workers across sectors and states.
- **A high-dimensional spatial model.** The paper allows for international trade, domestic interregional trade, intersectoral input-output linkages, local fixed factors, and labor mobility frictions in one framework.
- **Dynamic hat algebra.** The central technical contribution is showing that the model can be solved using observed allocations and changes, not the levels of hard-to-identify fundamentals. This makes a large-scale dynamic model computationally feasible.
- **Measurement discipline.** The model is matched to data on labor allocations, migration flows, trade flows, value added, input-output shares, and expenditure shares, rather than being solved around an assumed steady state.
- **Distributional welfare.** The paper calculates welfare changes for workers initially located in different state-sector markets, revealing large cross-market dispersion hidden by the positive aggregate gain.
- **Policy extensions.** The framework is extended to study Social Security Disability Insurance, persistent migration decisions, alternative preferences, and other model variants. This shows that the method can handle policy experiments beyond the China shock.

3 Data and measurement

3.1 Geography, sectors, and labor markets

- The model includes 50 U.S. states, 37 foreign countries, and a constructed rest of the world.
- There are 22 productive sectors: 12 manufacturing sectors, 8 service sectors, construction, and wholesale/retail trade.
- The paper excludes agriculture, mining, utilities, and the public sector from the core quantitative analysis.
- A U.S. labor market is a state-sector pair. Including non-employment, this produces 1,150 U.S. labor markets.

Interpretation: The disaggregation matters because the same aggregate China shock has different local consequences depending on state industry specialization and the local ability of workers to move.

3.2 Trade and production data

- International bilateral trade shares are constructed from the World Input-Output Database.
- U.S. interstate trade flows are built using the Commodity Flow Survey combined with WIOD data.
- State-level exposure to international trade is allocated using regional sectoral employment shares, following the logic that regions specialized in a sector are more exposed to shocks hitting that sector.

- Input-output coefficients, value-added shares, and labor shares are constructed from BEA data for U.S. states and WIOD data for other countries.
- Portfolio shares are chosen to match observed trade imbalances in the initial year.

Interpretation: The trade and production data are not just background inputs. They determine how cheaper Chinese goods affect each sector both directly through import competition and indirectly through cheaper intermediate inputs.

3.3 Labor mobility data

- The initial distribution of workers across states, sectors, and non-employment comes from the 2000 U.S. Census 5% Public Use Microdata Sample.
- The analysis focuses on people aged 25 to 65 who are either non-employed or employed in one of the model sectors.
- Quarterly mobility across sectors is constructed using CPS data, while interstate mobility is constructed using ACS data.
- The median quarterly probability of staying in the same state and sector is about 93.9%. The median probability of changing sector but not state is about 5.44%, while changing state is much rarer.

Interpretation: The mobility data imply large frictions. Workers can move, but they do not instantly jump to expanding sectors or states. This slow adjustment is central to the paper's welfare and transition-cost results.

3.4 Elasticities and calibrated parameters

- The quarterly discount factor is set to 0.99.
- Sectoral trade elasticities come from Caliendo and Parro (2015).
- The migration elasticity is estimated by adapting the Artuc-Chaudhuri-McLaren procedure to the quarterly frequency and log utility. The preferred estimate implies $\nu = 5.3436$.

Interpretation: The trade elasticities govern how strongly trade flows respond to costs and productivity. The migration elasticity governs how strongly workers respond to real-wage and option-value differences across labor markets.

3.5 Measuring the China trade shock

- The authors first estimate predicted U.S. imports from China by using growth in imports from China by other advanced economies.
- They then find Chinese sectoral productivity changes that make the model match the predicted change in U.S. imports from China from 2000 to 2007.
- The computer and electronics sector accounts for the largest share of the predicted import increase from China, followed by textiles and metals.

- The China shock can also be interpreted as an origin-specific trade-cost decline, because in the model's gravity structure a China-specific productivity increase and a China-specific trade-cost decline are isomorphic in many respects.

Interpretation: The paper does not simply feed observed imports into the model as exogenous. It isolates the part of U.S. import growth attributable to China-side changes, then maps that shock into the structural model.

4 Models and empirical specifications

4.1 Household dynamic decision problem

Workers begin a period in a region-sector market (n, j) . They receive current consumption from market wages if employed or home production if non-employed. They then choose where to be next period, paying mobility costs and receiving idiosyncratic preference shocks.

The expected value function has the logit form:

$$V_t^{nj} = U(C_t^{nj}) + \nu \log \left[\sum_i \sum_k \exp \left(\frac{\beta V_{t+1}^{ik} - \tau^{nj,ik}}{\nu} \right) \right].$$

The implied migration share from market (n, j) to (i, k) is:

$$\mu_t^{nj,ik} = \frac{\exp \left((\beta V_{t+1}^{ik} - \tau^{nj,ik}) / \nu \right)}{\sum_m \sum_h \exp \left((\beta V_{t+1}^{mh} - \tau^{nj,mh}) / \nu \right)}.$$

- Higher destination lifetime utility raises migration inflows.
- Higher origin-destination mobility costs reduce migration.
- A higher $1/\nu$ means migration is more responsive to utility differences.

Interpretation: The household side is the source of labor-market dynamics. Workers anticipate future conditions, but moving is costly and gradual.

4.2 Production, trade, and input-output linkages

- Firms produce intermediate goods with labor, local structures, and materials from all sectors.
- Productivity follows an Eaton-Kortum Frechet structure, producing gravity-style trade shares.
- Trade costs are iceberg costs between origins and destinations.
- Local sectoral aggregate goods are used both for final consumption and as intermediate inputs.
- Input-output linkages allow cheaper Chinese manufacturing inputs to raise production and labor demand in non-manufacturing sectors.

A representative sectoral trade share has the form:

$$\pi_t^{nj,ij} = \frac{(x_t^{ij} \kappa_t^{nj,ij})^{-\theta_j} (A_t^{ij})^{\theta_j \gamma^{ij}}}{\sum_m (x_t^{mj} \kappa_t^{nj,mj})^{-\theta_j} (A_t^{mj})^{\theta_j \gamma^{mj}}}.$$

Interpretation: Trade reallocates demand toward lower-cost sources. When Chinese productivity rises, U.S. manufacturing producers face direct competition, but U.S. firms and consumers also obtain cheaper goods and inputs.

4.3 Temporary and sequential equilibrium

- A **temporary equilibrium** solves wages, prices, expenditures, and trade shares at a given date, conditional on the distribution of labor and contemporaneous fundamentals.
- A **sequential equilibrium** adds the household migration decision and the law of motion for labor across all future dates.
- A **stationary equilibrium** is a special case in which allocations and fundamentals are constant over time.

Interpretation: The model separates the static production/trade block from the dynamic labor block. That separation is what makes the dynamic hat algebra tractable.

4.4 Dynamic hat algebra

The core methodological result is that the model can be solved in time differences. Let $\dot{y}_{t+1} = y_{t+1}/y_t$. Given observed allocations at date t , the authors express the temporary equilibrium at date $t + 1$ in changes.

- Productivity levels, trade-cost levels, migration-cost levels, local factors, and home-production levels do not need to be estimated.
- Observed allocations are sufficient statistics for the fundamentals needed to solve counterfactual changes.
- The dynamic system uses changes in migration shares, lifetime utilities, and labor allocations to solve the transition path.
- This is a dynamic extension of exact hat algebra used in static quantitative trade models.

Interpretation: This is the paper’s main technical innovation. It lets the authors study a high-dimensional dynamic model without separately identifying thousands of fundamentals.

4.5 Counterfactual China-shock experiment

The main counterfactual is:

Agents expected the actual evolution of all fundamentals from 2000 to 2007, but the China shock did not occur. What would have happened differently?

- The baseline economy follows actual allocations and actual inferred changes in fundamentals.
- The counterfactual economy removes the Chinese productivity changes that generated the predicted import surge.
- Differences between the baseline and counterfactual economies measure the effect of the China shock.

Interpretation: The comparison is not China shock versus a frozen economy. It is China shock versus an economy with all other observed changes still occurring.

4.6 Welfare and adjustment costs

The welfare change for workers initially in market (n, j) is measured as consumption equivalent variation:

$$\widehat{W}^{nj} = \sum_{s=1}^{\infty} \beta^s \log \left(\widehat{C}_s^{nj} (\widehat{\mu}_s^{nj, nj})^\nu \right).$$

- $\widehat{C}_s^{nj}$ captures real consumption changes.
- $\widehat{\mu}_s^{nj, nj}$ captures the option-value component from mobility decisions.
- Adjustment costs compare welfare when the new steady state is reached instantly with welfare along the actual transition path.

Interpretation: Welfare is not just real wages. It also includes the value of being able to move, but mobility frictions delay the realization of gains.

5 Identification and empirical design

5.1 What the model can identify

- The framework identifies the model-implied general equilibrium effects of a measured China-side shock, conditional on the structure of the model and the observed baseline allocations.
- The model can compute aggregate job effects, sectoral spillovers, regional effects, and welfare effects that reduced-form difference-in-differences regressions cannot directly deliver.
- The results are structural counterfactuals, not randomized estimates.

Interpretation: The paper is best read as a quantitative general equilibrium complement to reduced-form evidence, not a substitute for the reduced-form identification of local exposure effects.

5.2 Identification of the China shock

- The China shock is based on the predicted component of U.S. import growth from China.
- The predictor is import growth from China in other high-income countries, following the Autor-Dorn-Hanson identification logic.
- The paper then maps predicted import changes into Chinese sectoral productivity changes using the model.

Interpretation: This strategy aims to isolate China-side productivity or cost changes from U.S.-specific demand shocks.

5.3 Key mechanisms

- **Direct import competition:** U.S. manufacturing sectors exposed to China shrink.
- **Input-output effects:** Non-manufacturing sectors benefit from cheaper imported intermediate goods.
- **Spatial specialization:** States specialized in exposed industries lose more manufacturing jobs.
- **Worker mobility:** Workers gradually move toward expanding sectors and states, but adjustment is slow.
- **Local fixed factors:** Inflows and outflows affect local real wages through congestion and local factor prices.

Interpretation: The sign and size of a local effect depend on multiple layers of exposure, not just the import share of a single industry.

5.4 Limitations of the design

- The model assumes perfect foresight about the path of fundamentals.
- The core model abstracts from international migration.
- Domestic trade flows at the state-sector level must be constructed from multiple data sources and assumptions.
- The model does not directly estimate the levels of mobility costs or productivity; this is a strength for tractability, but it means the paper focuses on counterfactual changes rather than recovered fundamentals.
- The structural results depend on the correctness of the trade, production, and migration structure imposed by the model.

Interpretation: The paper’s power comes from combining a rich model with exact-difference methods. The cost is reliance on the maintained structure of the model.

6 Tables and figures

6.1 Table I and Table II: mobility and persistence

Read:

- Table I reports quarterly mobility moments. Most workers remain in the same state and sector. Sector switching is more common than interstate migration.
- The median probability of staying in the same state-sector market is 93.9%; the median probability of changing sector but not state is 5.44%.
- Table II shows robustness to additional persistence in migration decisions. As the persistence parameter ρ rises from 0 to 0.3, the manufacturing employment effect becomes slightly smaller, moving from -0.361 to -0.296 percentage points.

Economic message: Labor adjustment is slow and heterogeneous. The China shock has persistent effects because workers do not instantaneously move across sectors and states.

TABLE I
U.S. INTERSTATE AND INTERSECTORAL LABOR MOBILITY^a

Probability	p25	p50	p75
Changing sector but not state	3.58%	5.44%	7.93%
Changing state but not sector	0.04%	0.42%	0.73%
Changing state and sector	0.02%	0.03%	0.05%
Staying in the same state and sector	91.4%	93.9%	95.8%

^aQuarterly transitions. Data sources: ACS and CPS.

(a) Table I: U.S. interstate and intersectoral labor mobility

TABLE II
AGGREGATE EFFECTS ACROSS MODELS WITH
DIFFERENT DEGREES OF PERSISTENCE^a

Model		Change in Mfg. Emp. Share
ρ	ν	
0	5.3436	-0.361
0.1	5.0369	-0.315
0.2	4.6973	-0.302
0.3	4.3189	-0.296

^aThis table presents long-run employment effects due to the China shock, under different values of ρ and ν .

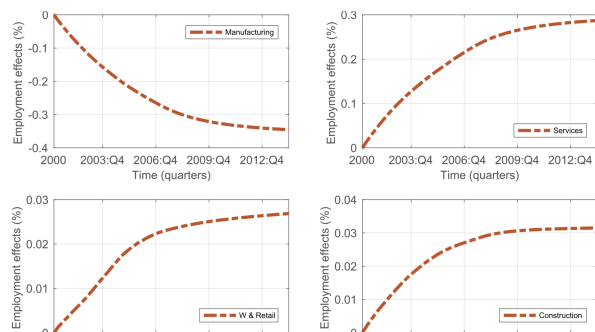
(b) Table II: aggregate effects with persistent migration decisions

6.2 Figures 1 and 2: aggregate sector dynamics and manufacturing contributors

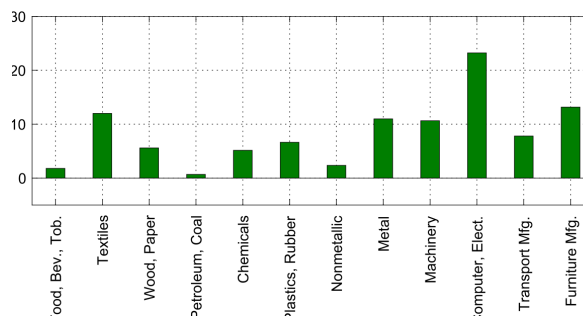
Read:

- Figure 1 shows that the China shock lowers the manufacturing employment share over time and raises employment shares in services, wholesale/retail, and construction.
- The manufacturing decline reaches about 0.36 percentage points in the long run.
- Figure 2 decomposes the manufacturing employment loss by industry. Computer and electronics contributes the largest share, roughly one-fourth of the total decline.

Economic message: The China shock is not just a manufacturing decline. It causes a reallocation of labor across the whole economy, with non-manufacturing sectors absorbing some displaced workers.



(a) Figure 1: employment-share effects over time



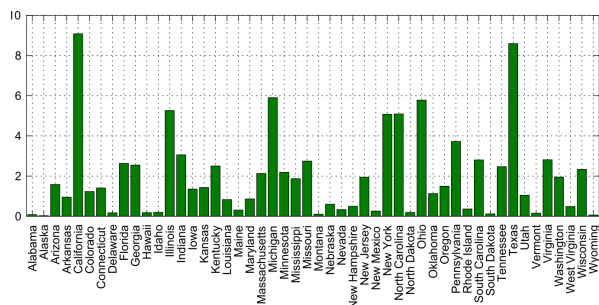
(b) Figure 2: manufacturing decline by industry

6.3 Figures 3 and 4: regional manufacturing exposure

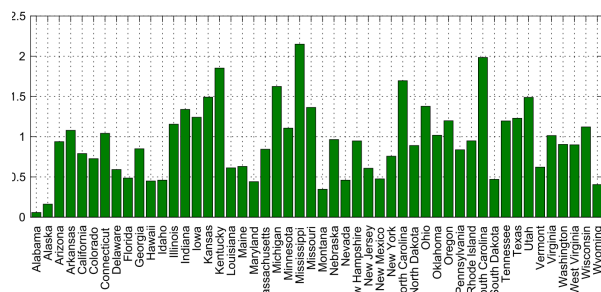
Read:

- Figure 3 shows each state's contribution to the aggregate U.S. manufacturing employment decline caused by the China shock.
- California and Texas contribute substantially because of their sectoral composition and size.
- Figure 4 normalizes by each state's employment share. Smaller manufacturing-intensive states such as South Carolina, Mississippi, and Kentucky show relatively large local impacts.

Economic message: Aggregate contributions and local pain are different concepts. Large states contribute more to national totals, while smaller manufacturing-intensive states can experience larger local effects.



(a) Figure 3: regional contribution to manufacturing decline



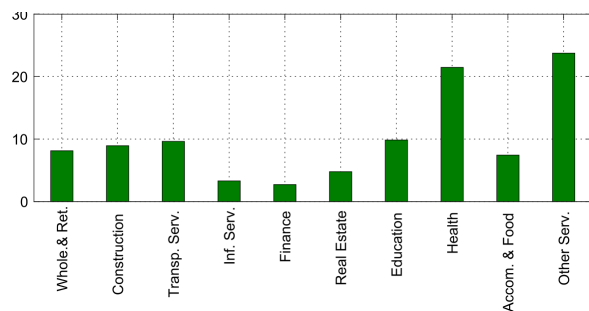
(b) Figure 4: contribution normalized by regional employment share

6.4 Figures 5 and 6: non-manufacturing absorption

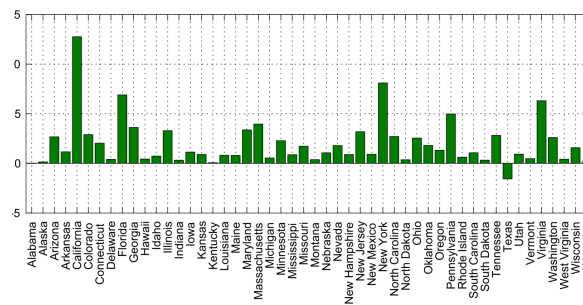
Read:

- Figure 5 shows that employment increases across non-manufacturing sectors, especially other services, health, education, construction, and transport services.
- Figure 6 shows the regional contribution to non-manufacturing employment increases. Large service-intensive states such as California and New York contribute heavily.

Economic message: Cheaper Chinese goods and intermediate inputs expand parts of the non-manufacturing economy. This is why a general equilibrium model delivers both job losses and job gains.



(a) Figure 5: non-manufacturing employment increase by industry



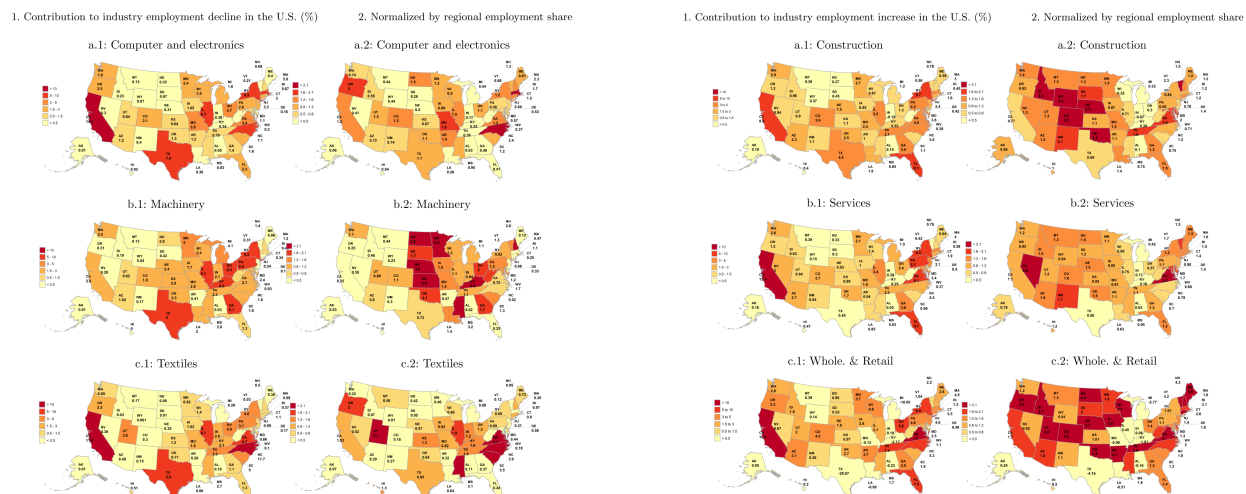
(b) Figure 6: regional contribution to non-manufacturing increase

6.5 Figures 7 and 8: selected industry-state maps

Read:

- Figure 7 maps employment declines in selected manufacturing industries: computer and electronics, machinery, and textiles.
- The computer/electronics decline is concentrated in California, while machinery losses are concentrated in the Midwest.
- Figure 8 maps employment increases in construction, services, and wholesale/retail. Non-manufacturing gains are more spatially diffuse than manufacturing losses.

Economic message: The China shock is transmitted through the geography of production. Manufacturing losses are localized in particular industry-state cells, while non-manufacturing gains are more broadly spread.



(a) Figure 7: regional manufacturing declines in selected industries

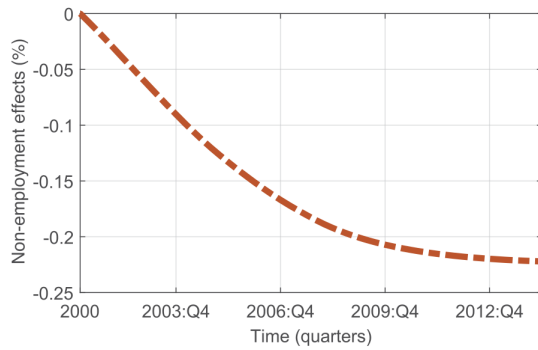
(b) Figure 8: regional non-manufacturing increases in selected industries

6.6 Figures 9 and 10: non-employment and welfare distribution

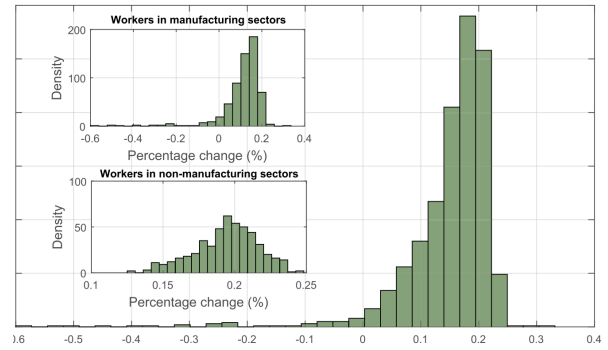
Read:

- Figure 9 shows that the model-implied aggregate non-employment share falls by about 0.22 percentage points in the long run.
- The decline in non-employment is driven mainly by expansion in non-manufacturing industries.
- Figure 10 shows a highly heterogeneous welfare distribution across 1,150 U.S. labor markets, with effects ranging from negative losses to gains near 1%.

Economic message: The average welfare gain hides distributional conflict. Some markets face losses or small gains, while many non-manufacturing markets benefit from cheaper goods and expanding labor demand.



(a) Figure 9: non-employment effects



(b) Figure 10: welfare effects across U.S. labor markets

6.7 Figures 11 and 12: long-run welfare versus short-run real wages

Read:

- Figure 11 maps long-run welfare effects across states. Aggregate welfare rises in all states, and gains are larger in non-manufacturing than in manufacturing.
- Figure 12 maps real-wage changes in manufacturing at impact and from 2000 to 2007. Manufacturing real wages fall in the short run and the decline deepens during the shock period.

Economic message: Long-run welfare gains and short-run wage losses can coexist. Trade gains materialize only after costly reallocation.

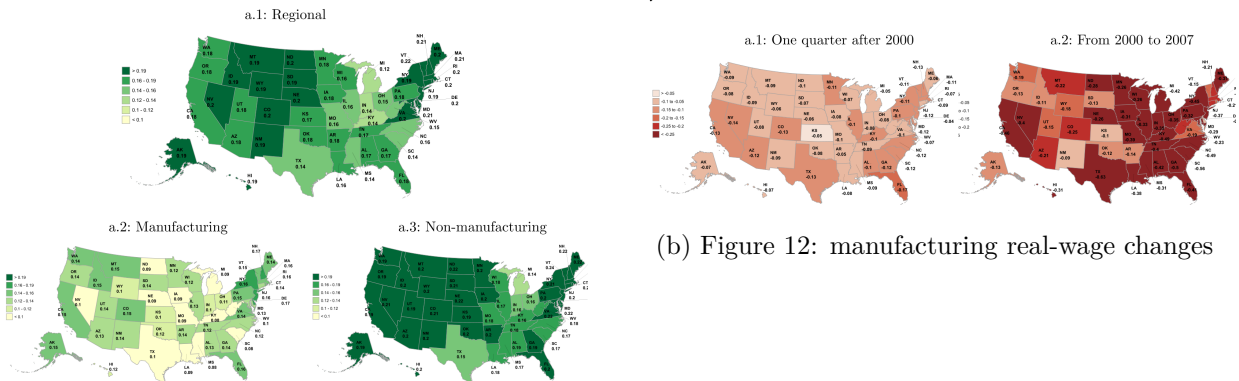


Figure 11: Regional welfare effects (percent)

(b) Figure 12: manufacturing real-wage changes

(a) Figure 11: regional welfare effects

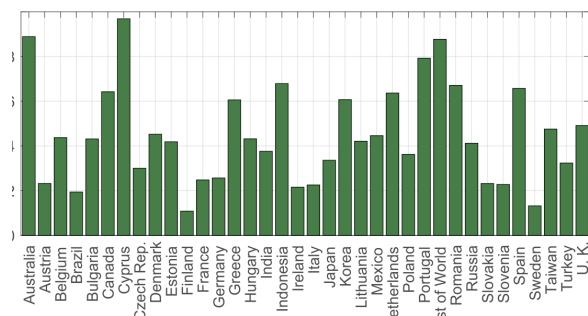
6.8 Figures 13 and 14: world welfare and adjustment costs

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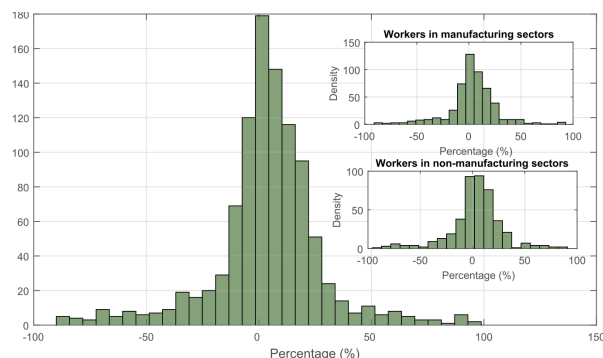
- Figure 13 shows that all sample countries gain from the China shock, though gains differ by openness and input-output exposure.
- Figure 14 shows adjustment costs across labor markets. The distribution has a long tail, meaning some labor markets face very large transition costs.

- The paper reports average transition costs of about 4.7% of steady-state welfare gains.

Economic message: China productivity growth is globally beneficial in the model, but adjustment frictions strongly affect who receives gains quickly and who bears transitional losses.



(a) Figure 13: welfare effects across countries



(b) Figure 14: adjustment costs across labor markets

6.9 Appendix Tables III and IV: measurement details

Read:

- Table III gives the concordance from SIC87dd sectors to the NAICS sectors used in the model.
- Table IV summarizes the information available from ACS and CPS that is used to construct the quarterly transition matrix across states, industries, and non-employment.

Economic message: The quantitative model requires substantial data construction. The appendix tables show how raw labor and industry data are mapped into the model's state-sector structure.

TABLE III
CONCORDANCE SIC87DD—NAICS^a

NAICS	NAICS Sector Description	SIC87dd Codes
1	Food, Beverage, and Tobacco Products	20**, 21**
2	Textiles and Apparel Products	22**, 23**, 31**
3	Wood, Paper, Printing and Related Products	24** exc. 241*, 26**, 274*-279*
4	Petroleum and Coal Products	29**
5	Chemicals	28**
6	Plastics and Rubber Products	30**
7	Nonmetallic Mineral Products	32**
8	Primary and Fabricated Metal Products	33**, 34**
9	Machinery	351*-356*, 3578-3599
10	Computer, Electrical, and Appliance	3571-3577, 365*-366*, 3812-3826, 3829, 386*-387*, 361*-364*, 367*-369*
11	Transportation Equipment	37**
12	Furniture and Miscellaneous Products	25**, 3827, 384*-385*, 39**
16	Information and Communication	271*-273*

^aAn entire broad group was mapped into the NAICS sector by substituting the last one or two digits with an asterisk. All intervals listed in the table are inclusive.

(a) Table III: SIC-to-NAICS concordance

TABLE IV
INFORMATION AVAILABLE ON ACS AND CPS

		State A				State B			
		Ind 1	Ind 2	...	Ind J	Ind 1	Ind 2	...	Ind J
State A	Ind 1	x	x	...	x				
	Ind 2	x	x	...	x				
	...	...	...	...	...				
	Ind J	x	x	...	x				
	Total	y	y	...	y	y	y	...	y
State B	Ind 1					x	x	...	x
	Ind 2					x	x	...	x
	...					...	...	...	...
	Ind J					x	x	...	x
	Total	y	y	...	y	y	y	...	y

(b) Table IV: information available from ACS and CPS

6.10 Appendix Table V and Figure 15: mobility validation and China shock measurement

Read:

- Table V compares actual and simulated mobility rates. The model closely matches quarterly sector switching and yearly state mobility.

- Figure 15 compares actual import changes, predicted import changes, and the implied sectoral Chinese TFP changes from 2000 to 2007.
- Computer and electronics is the largest import-growth category and has a large implied productivity increase.

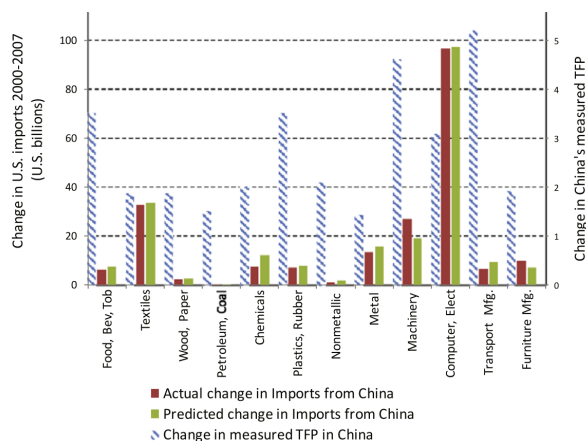
Economic message: The model’s mobility dynamics are plausible relative to the data, and the China shock is concentrated in specific manufacturing sectors rather than being uniform across all sectors.

TABLE V
ACTUAL AND SIMULATED MOBILITY RATES PERCENT^a

	Data	Model
Quarterly sector switching rate	6.1	5.4
Yearly state mobility rate	2.3	2.4

^a Model values are computed with simulated individual histories over 120 periods. Data on yearly state mobility rate computed using the ACS, 2001–2007. Data on quarterly sector mobility rates computed using the CPS, 2000–2007. Source: Author’s calculations.

(a) Table V: actual and simulated mobility rates



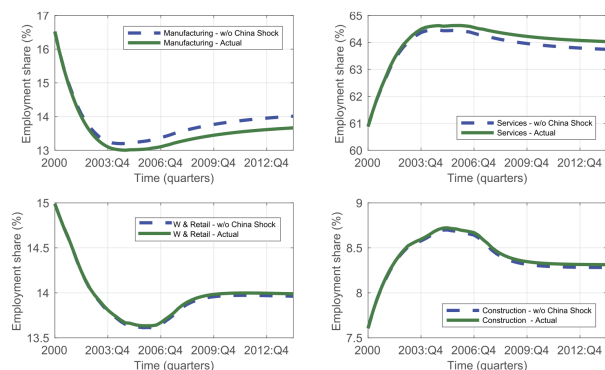
(b) Figure 15: imports and implied China TFP changes

6.11 Appendix Figures 16 and 17: baseline paths and additional manufacturing maps

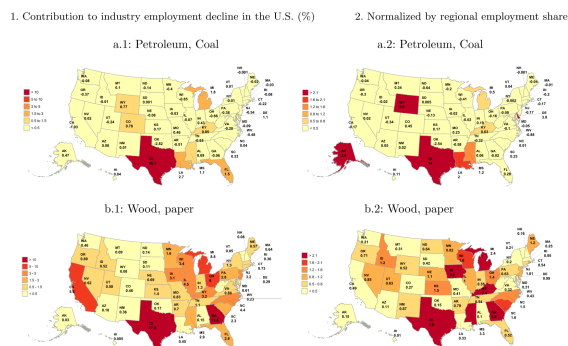
Read:

- Figure 16 compares actual sectoral employment shares with the no-China-shock counterfactual path.
- Figure 17 provides additional manufacturing maps for petroleum/coal and wood/paper.

Economic message: The appendix figures confirm that the model is matching observed broad employment movements while also showing the detailed geography of additional sectors.



(a) Figure 16: evolution of employment shares



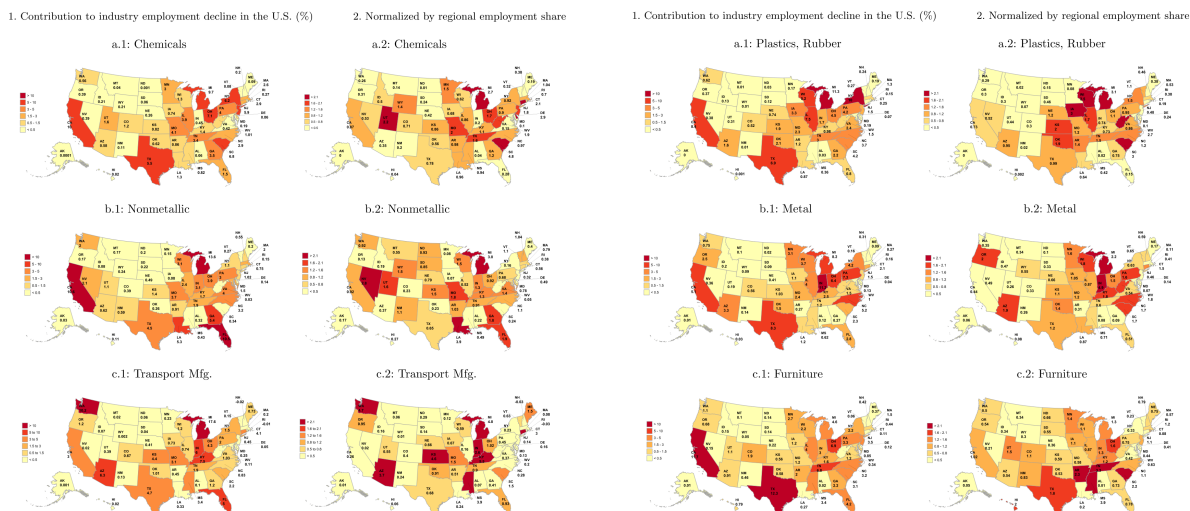
(b) Figure 17: additional manufacturing maps

6.12 Appendix Figures 18 and 19: additional manufacturing heterogeneity

Read:

- Figure 18 maps chemicals, nonmetallic products, and transportation manufacturing.
- Figure 19 maps plastics/rubber, metal, and furniture.
- These maps show that manufacturing losses differ strongly by state and industry.

Economic message: There is no single “manufacturing region.” The China shock operates through many specific state-industry cells.



(a) Figure 18: chemicals, nonmetallic, and transport manufacturing

(b) Figure 19: plastics/rubber, metal, and furniture

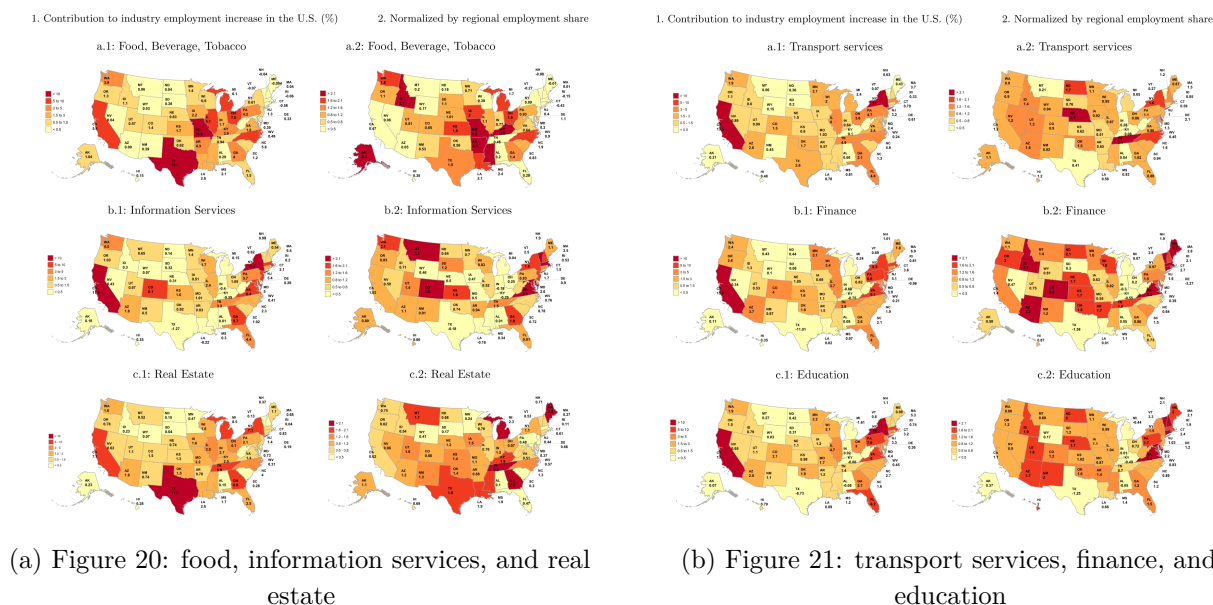
6.13 Appendix Figures 20 and 21: additional non-manufacturing gains

Read:

- Figure 20 maps non-manufacturing gains in food/beverage/tobacco, information services, and real estate.

- Figure 21 maps transport services, finance, and education.
- Compared with manufacturing losses, these gains tend to be less concentrated across states.

Economic message: Non-manufacturing sectors absorb workers in many parts of the country, not only in a few high-exposure manufacturing regions.



6.14 Appendix Figure 22: remaining non-manufacturing sectors

Read:

- Figure 22 maps health care, accommodation/food, and other services.
- These sectors help explain the broad-based non-manufacturing employment increase in the model.

Economic message: The service-sector response is broad. The employment gains from the China shock are not confined to a single non-manufacturing industry.

1. Contribution to industry employment increase in the U.S. (%) 2. Normalized by regional employment share

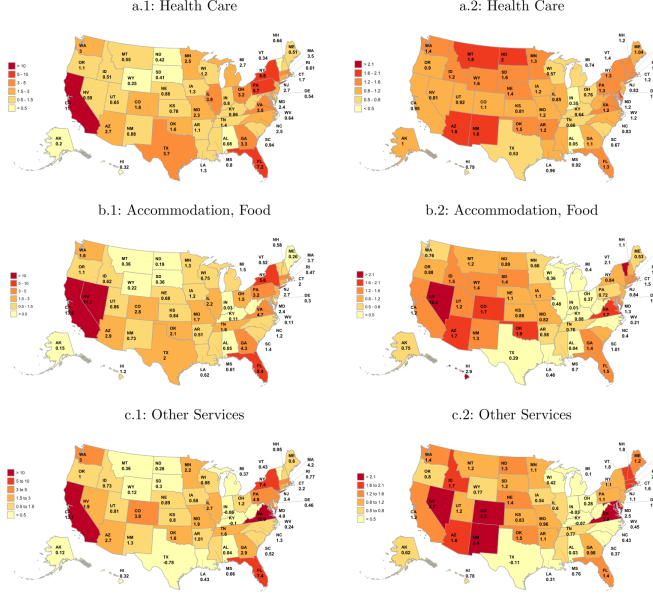


Figure 22: health care, accommodation/food, and other services

7 Short summary

- The paper builds a dynamic spatial trade and migration model to study the China trade shock.
- The model combines international trade, interstate trade, input-output linkages, local fixed factors, and costly labor reallocation.
- The methodological innovation is dynamic hat algebra, which solves the model in changes without estimating the levels of many hard-to-identify fundamentals.
- The empirical application covers 38 countries, 50 U.S. states, 22 sectors, and 1,150 U.S. labor markets.
- The China shock is measured using an Autor-Dorn-Hanson-style predicted import increase and mapped into Chinese sectoral productivity changes.
- The China shock reduces U.S. manufacturing employment by about 0.55 million jobs in the long run.
- The largest manufacturing losses come from computer and electronics, furniture, textiles, metal, and machinery.
- Some non-manufacturing sectors expand because they benefit from cheaper imported intermediate inputs.
- Aggregate U.S. welfare rises by about 0.2%, but welfare effects vary substantially across labor markets.
- Mobility frictions create short-run manufacturing real-wage losses and heterogeneous adjustment costs.

- The SSDI extension suggests disability benefits amplified the manufacturing decline by about 50,000 jobs.

8 Conclusion

8.1 Core conclusion

Caliendo, Dvorkin, and Parro show that the China trade shock reduced U.S. manufacturing employment but increased aggregate U.S. welfare. The central reason is that trade creates both losses and gains: import competition hurts exposed manufacturing markets, while cheaper goods and intermediate inputs benefit consumers and many non-manufacturing producers. Because workers face mobility frictions, the gains take time to materialize and the short-run costs are concentrated in particular labor markets.

8.2 Economic intuition

The key intuition is that trade shocks are not absorbed in one market. A productivity increase in China changes import prices, sectoral demand, intermediate-input costs, wages, local prices, migration incentives, and future option values. A worker in a manufacturing sector exposed to China may face lower local labor demand in the short run. But over time, some workers move to services, construction, or wholesale/retail, and firms outside manufacturing benefit from cheaper inputs. The final effect depends on the interaction of trade exposure, input-output linkages, geography, and mobility costs.

8.3 Incremental contribution revisited

The paper advances the China-shock literature in three ways. First, it moves from differential local exposure estimates to aggregate and distributional general equilibrium effects. Second, it incorporates dynamic worker reallocation rather than comparing static equilibria. Third, it provides a practical method - dynamic hat algebra - that makes a large dynamic spatial model empirically feasible without estimating all fundamentals.

8.4 Implications

For trade policy, the paper shows why aggregate welfare gains can coexist with severe localized labor-market losses. For labor-market policy, it highlights the importance of mobility frictions and adjustment costs. For empirical work, it provides a bridge between reduced-form exposure designs and structural counterfactuals. For policy counterfactuals, the framework can be extended to analyze disability insurance, unemployment benefits, migration costs, regional policies, and trade-cost changes.

8.5 Limitations

The results rely on perfect foresight, constructed trade and migration data, and a specific general equilibrium structure. The model abstracts from international migration in the quantitative application and does not directly estimate the levels of productivity, trade costs, or migration costs.

The China shock is measured through a particular ADH-style import-prediction procedure. These choices are reasonable and disciplined, but they shape the quantitative conclusions.

8.6 Takeaway

The enduring takeaway is that the China trade shock cannot be understood only as a decline in manufacturing jobs. It is a dynamic, spatial, general equilibrium reallocation episode. The U.S. gains in the aggregate, but workers in different state-sector markets experience very different employment, wage, welfare, and transition-cost effects. The paper's contribution is to make those distributional dynamics quantitatively visible.